# Hassan Imani

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## PROFESSIONAL EXPERIENCE

**Data Scientist** | AML AI Research & Special Projects — Jul 2025 – Present
*Scotiabank* — *Toronto, ON*

- **Generative Anomaly Detection:** Optimized, stabilized, and refactored the team's unsupervised conditional generative model to isolate peer-group financial anomalies.
- **Optimization & Experimentation:** Developed an automated ensembling tool and designed proxy evaluation metrics to factor in anomalies flagged by different configurations of the model.
- **Transaction Alert Pipelines:** Engineered and deployed multiple end-to-end risk-specific transaction monitoring pipelines using PySpark, SQL, and PyTorch to detect financial compliance risk typologies.
- **Synthetic Simulation:** Built probabilistic generative models simulating privacy-preserving transactional behaviors, eliminating cold-start and validation bottlenecks for benchmarking unsupervised anomaly models.
- **Representation Learning:** Validated the semantic consistency (KYC descriptors) of client embeddings generated by self-supervised sequence models processing high-volume transactions.

**Machine Learning Engineer** — Feb 2024 – Jul 2025
*Zown* — *Toronto, ON*

- **Two-Tower Recommender:** Built a multi-stage property matching engine utilizing hard-constraint filtering followed by a neural Two-Tower architecture; integrated auxiliary multi-task objectives (dwell time, engagement) to enrich embeddings.
- **MLOps & Pipelines:** Implemented an exploration-set strategy to solve user cold-start and automated offline micro-batch retraining gated by holdout-set validation to prevent production metric degradation.
- **In-House Computer Vision:** Fine-tuned a Vision Transformer (Google ViT) with multi-task classification and regression heads to extract listing photo attributes, eliminating external API dependencies.
- **AI Valuation & Re-targeting:** Developed a deep tabular model with a Continuous Training (CT) pipeline for property valuation; designed lead-conversion analyses using clustering and Positive-Unlabeled (PU) learning to re-target hot cohorts.

## EDUCATION

**York University** — 2021 - 2025
*PhD in Mechanical Engineering* — *Toronto, ON*

- Specialized in spatio-temporal tracking, GNNs, and statistical regression.

**Tehran Polytechnic** — 2018 - 2021
*MSc in Mechanical Engineering* — *Tehran, IR*

**Sharif University of Technology** — 2013 - 2018
*BSc in Mechanical Engineering* — *Tehran, IR*

## TECHNICAL SKILLS

**Languages & Distributed**: Python, SQL, Bash, PySpark, Spark

**ML & Statistics**: Anomaly Detection, Representation Learning, Recommender Systems, Multi-task Learning, Vision Transformers

**Tools & Platforms**: Git, Docker, GCP, MongoDB, PyTorch, Copilot

## PUBLICATIONS

Graph Neural Networks for Sustainable Energy (HI Parashkooh, C Jian) — 2025
*Predicting Adsorption in Aromatic Molecules* — *ChemE 9 (4), 85*

An Innovative Method for Calculating PMV (HI Parashkooh, C Jian) — 2024
*from a standard molecular trajectory* — *J. Mol. Liq. 395, 123879*

Data Mining Guided Molecular Investigations (HI Parashkooh, C Jian) — 2022
*on the Coalescence of Water-in-Oil Droplets* — *Energy Fuels 36 (4), 1811*